

# Positioning of OmniGene-4-MM relative to recent MoE bio-models

## AIDO.Protein Sun et al. 2024

### Protein only

- 16B-A8B sparse MoE, trained from scratch
- 1.2 trillion tokens
- 256 × A100, 64 days
- $\approx 16,384$  GPU-days
- Goal: SOTA on PEER / ProteinGym

≡ specialized tool

## Tripathi et al. 2025 Sci Reports

### DNA only

- MoE-of-CNN ensemble (DeepBIND backbone)
- Task: TFBS binary classification
- ShiftSmooth XAI attribution
- $\approx 10$  GPU-days
- Goal: stable TFBS classifier

≡ specialized tool

## OmniGene-4-MM (this work)

### Vision · Sequence · Language 8 modalities total

- Gemma-4 26B-A4B MoE + LoRA  $r=64$
- $\approx 100$  M tokens (PAWS-X paraphrase)
- Single H20, 30 h  $\approx 1.25$  GPU-days
- Modality-invariant syntactic transfer
- Router-level XAI (3-tier emergence)

≡ **mechanism + tool**

Training compute (log scale, GPU-days)

16,384 GPU-days

10 GPU-days

1.25 GPU-days